Meeting and Presentation Plan

A shareable agenda, demonstration flow and four-person speaking guide

STAT7006A | Spatial Statistics Dashboard | Final bright-theme candidate

1. What the group needs to decide

This is a final candidate, not a declaration that the group has already chosen it. Use the meeting to decide whether to adopt this dashboard as the common group design, or whether any remaining improvements are essential before presentation.

Decision	Suggested approach
Dashboard version	Compare final candidate with teammates' alternatives; agree on one version and one owner for deployment.
Data consistency	Retain the established NOAA-21 2024–2025 dataset unless the group jointly approves a documented change.
Features	Preserve the seven tabs, national/provincial analysis, interpretations and seasonal comparisons.
Evidence	Confirm provincial boundary provenance and any result examples used on slides.
Responsibility	Assign one lead for the app, one for data/methods, one for results and one for limitations/Q&A.
Submission	Confirm exact course submission instructions, GitHub repository contents and deployed app URL.

2. Suggested 20-minute group meeting agenda

Minutes	Activity
0–3	Open the live dashboard and confirm all group members can access it.
3–8	Walk through Start Here, Overview and Seasonal Fire Intelligence.
8–13	Run a provincial KDE and NN analysis; show automatic interpretation; discuss national option.
13–16	Compare this candidate with any alternative design and agree what to keep.
16–20	Assign speaking roles, GitHub responsibilities and rehearsal deadline.

3. Suggested four-person presentation division

Speaker	Focus	Sample speaking points
Member 1	Context, aim, data	"We use NASA FIRMS VIIRS NOAA-21 detections from 2024 and 2025 to examine where and when thermal anomalies occur across South Africa."
Member 2	Dashboard navigation and seasonal	"The filters let users explore one

	results	province or the whole country. The seasonal tab compares both years and reveals how detection counts vary by month and season."
Member 3	KDE and NN methods	"KDE maps concentration; nearest-neighbour analysis tests whether points are closer together than expected under a uniform random pattern."
Member 4	Ripley's K, limitations, wrap-up	"Ripley's K explores patterns at several distances. We show simulation envelopes, but they are exploratory and not a global significance test."

Adjust speaker allocation to actual group preferences. Avoid assigning every computationally intensive live demonstration to the same speaker without rehearsing transitions.

4. Live demonstration sequence (suggested 8–10 minutes)

1. Open Start Here: introduce the South African context, aims and dataset meaning.
2. Open Overview: select a province and year, show detection totals and map. Explain that dots represent detections.
3. Open Seasonal Fire Intelligence: compare 2024 with 2025; show monthly and seasonal charts. Do not claim raw counts equal wildfire incidence.
4. Open KDE: choose a reasonable bandwidth (e.g., 10 km) and run analysis; explain where detections concentrate and why bandwidth matters.
5. Open Nearest Neighbour: run a provincial example; explain observed/expected distance, ratio and the one-sided Monte Carlo p-value.
6. Open Ripley's K: show L(r)−r and pointwise CSR envelope; explain distance-dependent clustering and limitations.
7. Select All Provinces briefly to demonstrate national capability (prefer KDE for a quick live demo); mention sample limits.
8. Finish on Methods & Limitations and give the live URL / GitHub submission location.

5. Interpretation lines you can say aloud

Output	Plain-language explanation
KDE	"The warmest parts of this map are where satellite fire detections are most concentrated after smoothing. The result depends on the bandwidth we choose."
NN ratio below 1	"On average, the detections are closer together than the simple random-pattern benchmark would predict."
Small NN p-value	"Under the uniform random reference model, this much clustering would be unusual. It does not tell us what caused the pattern."
Ripley's L(r)−r above 0	"At this distance, we see more nearby pairs than the uniform-random reference expects."
Ripley's curve outside band	"This is an exploratory pointwise comparison with simulations, not a global significance test."

Year-on-year change	“The satellite recorded more or fewer detections, but we cannot directly infer a change in the number of independent fires or burned area.”

6. Likely questions and concise answers

Why NOAA-21 rather than another sensor?

It is the source used consistently in this project. Combining different sensors would require harmonization and a documented comparison of detection characteristics.

Are detections the same as fires?

No. A single fire can produce repeated detections and some fires may be missed.

Why three methods?

KDE describes intensity, NN tests overall short-distance proximity relative to a reference model, and Ripley’s K examines clustering across multiple distances.

What does $p < 0.05$ mean here?

For the NN one-sided Monte Carlo test, the observed mean distance is unusually short under the specified uniform-CSR simulation model. It does not establish causation.

Is Ripley’s K statistically significant?

The displayed 19-simulation envelopes are pointwise exploratory references. The app does not compute a formal global envelope test.

Can the app analyse the whole country?

Yes. All Provinces enables national analysis using a South Africa-centred LAEA projection and reproducible point samples.

Why sample the points?

To keep interactive calculations responsive on limited hosting resources; results should be understood as sample-based.

What are the main limitations?

Repeated satellite detections, unadjusted counts, variable detection opportunity, a simplified CSR model, and scale/projection effects.

What happens if the app disconnects?

Refresh the hosted page, wait for restart, and continue from the last stable demonstration point; have screenshots or notes as backup.

7. Final presentation-day checklist

- Agree on the final version and avoid last-minute untested code changes.
- Check live URL: https://brudance.shinyapps.io/firewatch-sa/
- Keep the same app.R and both RDS files in the deployment folder.
- Test all seven tabs, at least one download and all three statistical methods.
- Open the dashboard a few minutes before presenting; avoid leaving it idle for long.
- Have screenshots of results and a PDF of speaker notes available as fallback.
- Confirm GitHub submission requirements and avoid uploading API tokens or secrets.
- Confirm the provincial boundary data citation before submitting documentation.

Final selected version (23 October 2026): The KDE panel now displays an interactive OpenStreetMap basemap beneath the density raster. A separate static PNG download is retained. See Document 04 for implementation details, installation, testing, and caveats. This addition supersedes earlier descriptions of a KDE plot without a basemap.